

Experiment Details

We conduct eight controlled experiments in a local ABIDES-based continuous double auction environment. ABIDES is used because it is a discrete-event market simulator designed for AI market research, supports pairwise latency, and uses message-based exchange communication. ABIDES-Gym can be used as a convenience wrapper for repeated runs and structured experimentation.

To keep market size constant across conditions, each run contains **10 total agents**. For Experiments 1–7, the market consists of **1 LLM trader** and **9 traditional agents**. The traditional agents provide a stable benchmark population and may include a mixture of ZIC-style, ZIP-style, and simple liquidity-providing or liquidity-taking strategies. For Experiment 8, we replace the single LLM trader with **2 communicating LLM traders**, while the remaining **8 agents** are traditional agents. This preserves the same total number of agents while allowing direct study of language-based interaction.

Experiment 1: Baseline market reproduction

Run ABIDES with only traditional agents to verify that the market produces sensible bid-ask spreads, volume, and price discovery. This confirms that the simulator, matching logic, and logging pipeline are working correctly.

Experiment 2: Single LLM trader vs traditional background

Insert one LLM trader into a market with 9 traditional agents. Compare profit, fill rate, quote aggressiveness, and market impact against the same market without the LLM.

Experiment 3: Prompt ablation

Use the same 1-LLM + 9-traditional setup, but vary only the prompt format:

- minimal prompt,
- reasoning prompt,
- structured JSON prompt.

This tests whether more explicit reasoning improves behavior or simply adds latency and token cost.

Experiment 4: Memory ablation

Keep the same market composition and vary only the amount of historical information provided to the LLM:

- no history,
- last 5 events,

- last 20 events,
- rolling session summary.

This tests whether memory improves adaptation after shocks.

Experiment 5: Information-access ablation

Vary the market view given to the LLM:

- top-of-book only,
- full visible limit-order-book snapshot,
- limit-order-book plus recent trades,
- limit-order-book plus inventory and recent PnL.

This isolates the effect of information richness on trading quality.

Experiment 6: Latency sensitivity

Use the same market setup but add artificial delay to the LLM decision process:

- no delay,
- small delay,
- moderate delay,
- large delay.

This evaluates whether the LLM remains competitive when it loses reaction-speed advantages.

Experiment 7: Shock-response market

Introduce a sudden change in the fundamental value or supply-demand regime during the session. Measure how quickly the LLM adapts, how much profit it loses during the shock, and how long it takes quotes to re-center.

Experiment 8: Communication / collusion

Use **2 LLM traders** and **8 traditional agents**. The two LLMs are placed on the same side of the market, and the key treatment is whether they are allowed to exchange private messages before quoting. Compare:

- no communication,
- restricted communication,
- free communication.

Measure spread widening, undercutting frequency, quote synchronization, and seller-side surplus. This experiment is distinct from the first seven because communication requires at least two LLMs; a single LLM cannot meaningfully test inter-agent language coordination.